Crime Analysis

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1 Crime Incidents Reports Data

```
path <- here::here("SoSe_2026/databases/Exercises/")
crime_desc <- readxl::read_xlsx(file.path(path, "FieldExplanation.xlsx"))

con <- dbConnect(RSQLite::SQLite(), file.path(path, "crime_db.sqlite"))
crime_data <- tbl(con, "tbl_crime")
```

```
head(crime_data) %>% gt()
```

The dataset contains NA incident reports described using 17 variables.

```
crime_desc %>%
  separate(col = `Field Name, Data Type, Required`,
           into = str_split(names(crime_desc)[1],
                            pattern = ",")[[1]], sep = ",") %>%
  mutate("Field Name" = str_replace_all(`Field Name`,
                                        pattern = "[\\[\\]]",
                                        replacement = "")) %>%
  gt() %>%
  tab_caption("Test")
```

2 Explorative Analysis

2.1 Incidents by Day of the Week

INCIDENT_NUMBER	OFFENSE_CODE	OFFENSE_CODE_GROUP	OFFENSE_DESCRIPTIO
232007173	3115	NA	INVESTIGATE PERSON
232004454	3301	NA	VERBAL DISPUTE
232006290	3115	NA	INVESTIGATE PERSON
232024939	3114	NA	INVESTIGATE PROPER
232006708	423	NA	ASSAULT - AGGRAVAT
232007629	3115	NA	INVESTIGATE PERSON

Table 1: Test

Field Name	Data Type	Required	Description
incident_num	[varchar](20)	NOT NULL	Internal BPD report number
offense_code	[varchar](25)	NULL	Numerical code of offense description
Offense_Code_Group_Description	[varchar](80)	NULL	Internal categorization of [offense_de
Offense_Description	[varchar](80)	NULL	Primary descriptor of incident
district	[varchar](10)	NULL	What district the crime was reported
reporting_area	[varchar](10)	NULL	RA number associated with the wher
shooting	[char] (1)	NULL	Indicated a shooting took place.
occurred_on	[datetime2](7)	NULL	Earliest date and time the incident c
UCR_Part	[varchar](25)	NULL	Universal Crime Reporting Part num
street	[varchar](50)	NULL	Street name the incident took place

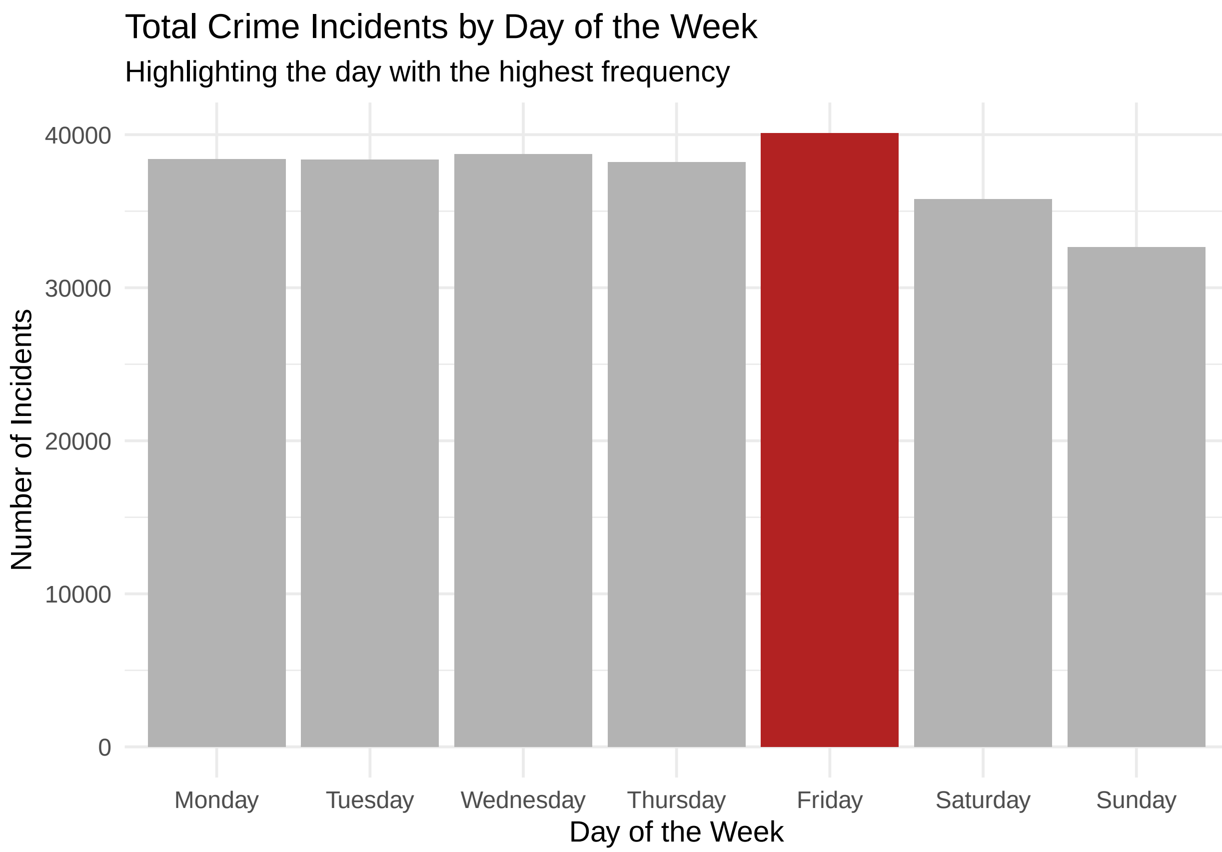
```r
weekdays_obj <- c("Sunday", "Monday", "Tuesday",
                  "Wednesday", "Thursday", "Friday", "Saturday")
weekly_stats <- crime_data %>%
  group_by(DAY_OF_WEEK) %>%
  summarise(incident_count = n()) %>%
  collect() %>%
  mutate(
    DAY_OF_WEEK = factor(DAY_OF_WEEK, levels = c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday"
    is_max = incident_count == max(incident_count)
  )
```

```r
ggplot(weekly_stats, aes(x = DAY_OF_WEEK, y = incident_count, fill = is_max)) +
  geom_col() +
  scale_fill_manual(values = c("gray70", "firebrick"), guide = "none") +
  labs(title = "Total Crime Incidents by Day of the Week",
       subtitle = "Highlighting the day with the highest frequency",
       x = "Day of the Week",
       y = "Number of Incidents") +
  theme_minimal()
```

Table 2: The data shows that Friday typically records the highest volume of incidents.

Weekly Crime Distribution

Day	Total Reports
Friday	40097
Monday	38401
Saturday	35778
Sunday	32639
Thursday	38211
Tuesday	38370
Wednesday	38717




```r
weekly_stats %>%
  select(DAY_OF_WEEK, incident_count) %>%
  gt() %>%
  tab_header(title = "Weekly Crime Distribution") %>%
  tab_caption(caption = "The data shows that Friday typically records
            the highest volume of incidents.") %>%
  cols_label(DAY_OF_WEEK = "Day", incident_count = "Total Reports")
```

2.2 Year-over-Year Analysis

```
incidents_by_month <- crime_data %>%
  summarise(Incidents = n(), .by = c(YEAR, MONTH)) %>%
  collect() %>%
  mutate(Month = lubridate::make_date(YEAR, MONTH, 1))

gg <- incidents_by_month %>%
  ggplot(aes(x = Month, y = Incidents)) +
    geom_line(col = "darkblue") +
  geom_point(col = "darkblue") +
  labs(title = "Incidents by Month") +
  theme_minimal()

if (knitr::is_html_output()) {
  plotly::ggplotly(gg)
} else if (knitr::is_latex_output()) {
  print(gg)
}
```